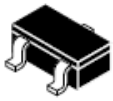
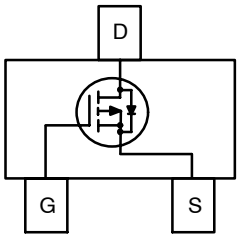
MOSFET – P-Channel, 1.8 V Specified, POWERTRENCH®

FDN306P



SOT–23
CASE 527AG



General Description

This P–Channel 1.8 V specified MOSFET uses **onsemi**'s advanced low voltage POWERTRENCH process. It has been optimized for battery power management applications.

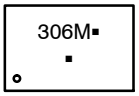
Features

- –2.6 A, –12 V $R_{DS(on)}$ = 40 mΩ @ V_{GS} = –4.5 V
  $R_{DS(on)}$ = 50 mΩ @ V_{GS} = –2.5 V
  $R_{DS(on)}$ = 80 mΩ @ V_{GS} = –1.8 V
- Fast Switching Speed
- High Performance Trench Technology for Extremely Low $R_{DS(on)}$
- SUPERSOT™–3 Provides Low $R_{DS(on)}$ and 30% Higher Power Handling Capability than SOT–23 in the Same Footprint
- This is a Pb–Free and Halide Free Device

Applications

- Battery Management
- Load Switch
- Battery Protection

MARKING DIAGRAM



306 = Specific Device Code
M = Month Code
▪ = Pb–Free Package

(Note: Microdot may be in either location)

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Value	Unit
V_{DSS}	Drain–Source Voltage	–12	V
V_{GSS}	Gate–Source Voltage	±8	V
I_D	Drain Current – Continuous (Note 1a) – Pulsed	 –2.6 –10	A
P_D	Maximum Power Dissipation (Note 1a) (Note 1b)	 0.5 0.46	W
T_J, T_{STG}	Operating and Storage Junction Temperature Range	–55 to +150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS (T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Thermal Resistance, Junction–to–Ambient (Note 1a)	250	°C/W
$R_{\theta JC}$	Thermal Resistance, Junction–to–Case (Note 1)	75	°C/W

ORDERING INFORMATION

Device	Package	Shipping†
FDN306P	SOT–23 (Pb–Free/ Halide Free)	3000 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

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ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
OFF CHARACTERISTICS						
BV_{DSS}	Drain–Source Breakdown Voltage	V_{GS} = 0 V, I_D = –250 μA	–12	–	–	V
ΔBV_{DSS} / ΔT_J	Breakdown Voltage Temperature Coefficient	I_D = –250 μA, Referenced to 25°C	–	–3	–	mV/°C
I_{DSS}	Zero Gate Voltage Drain Current	V_{DS} = –10 V, V_{GS} = 0 V	–	–	–1	μA
I_{GSSF}	Gate–Body Leakage, Forward	V_{GS} = 8 V, V_{DS} = 0 V	–	–	100	nA
I_{GSSR}	Gate–Body Leakage, Reverse	V_{GS} = –8 V, V_{DS} = 0 V	–	–	–100	nA
ON CHARACTERISTICS						
$V_{GS(th)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = –250 μA	–0.4	–0.6	–1.5	V
$\Delta V_{GS(th)}$ / ΔT_J	Gate Threshold Voltage Temperature Coefficient	I_D = –250 μA, Referenced to 25°C	–	2.5	–	mV/°C
$R_{DS(on)}$	Static Drain–Source On–Resistance	V_{GS} = –4.5 V, I_D = –2.6 A V_{GS} = –2.5 V, I_D = –2.3 A V_{GS} = –1.8 V, I_D = –1.8 A V_{GS} = –4.5 V, I_D = –2.6 A, T_J = 125°C	– – – –	30 39 54 40	40 50 80 54	mΩ
$I_{D(on)}$	On–State Drain Current	V_{GS} = –4.5 V, V_{DS} = –5 V	–10	–	–	A
g_{FS}	Forward Transconductance	V_{DS} = –5 V, I_D = –2.6 A	–	10	–	S
DYNAMIC CHARACTERISTICS						
C_{iss}	Input Capacitance	V_{DS} = –6 V, V_{GS} = 0 V, f = 1.0 MHz	–	1138	–	pF
C_{oss}	Output Capacitance		–	454	–	pF
C_{rss}	Reverse Transfer Capacitance		–	302	–	pF
SWITCHING CHARACTERISTICS (Note 2)						
$t_{d(on)}$	Turn–On Delay Time	V_{DD} = –6 V, I_D = –1 A, V_{GS} = –4.5 V, R_{GEN} = 6 Ω	–	11	20	ns
t_r	Turn–On Rise Time		–	10	20	ns
$t_{d(off)}$	Turn–Off Delay Time		–	38	61	ns
t_f	Turn–Off Fall Time		–	35	56	ns
Q_g	Total Gate Charge	V_{DS} = –6 V, I_D = –2.6 A, V_{GS} = –4.5 V	–	12	17	nC
Q_{gs}	Gate–Source Charge		–	2	–	nC
Q_{gd}	Gate–Drain Charge		–	3	–	nC
DRAIN–SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS						
I_S	Maximum Continuous Drain–Source Diode Forward Current		–	–	–0.42	A
V_{SD}	Drain–Source Diode Forward Voltage	V_{GS} = 0 V, I_S = –0.42 (Note 2)	–	–0.6	–1.2	V

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

NOTES:

1. $R_{\theta JA}$ is the sum of the junction–to–case and case–to–ambient thermal resistance where the case thermal reference is defined as the solder mounting surface of the drain pins. $R_{\theta JC}$ is guaranteed by design while $R_{\theta CA}$ is determined by the user's board design.

a) 250°C/W when mounted on a 0.02 in^2 pad of 2 oz. copper.

b) 270°C/W when mounted on a minimum pad.

Scale 1:1 on letter size paper.

2. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

TYPICAL CHARACTERISTICS

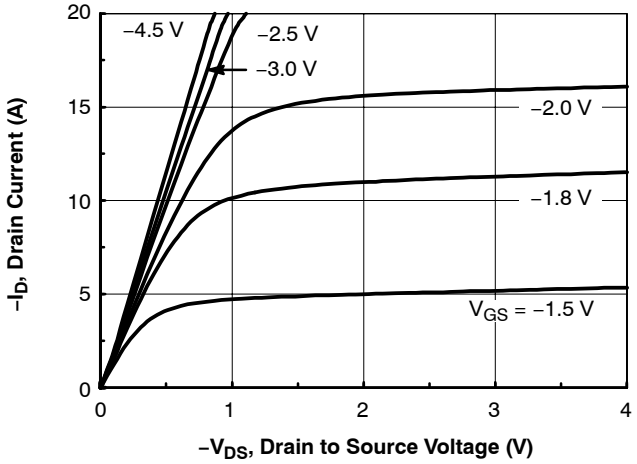

Figure 1. On−Region Characteristics

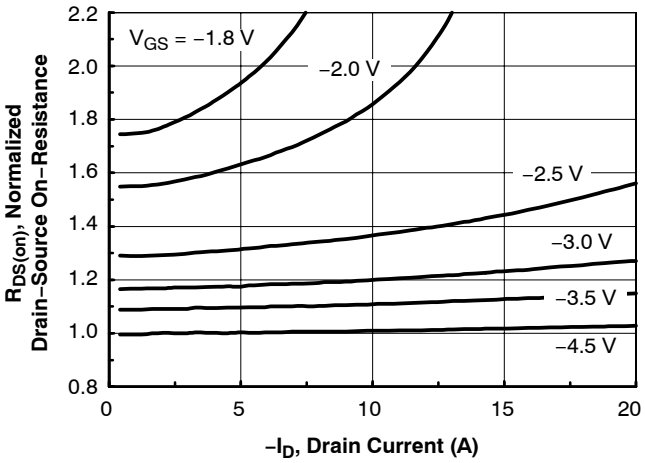

Figure 2. On−Resistance Variation with Drain Current and Gate Voltage

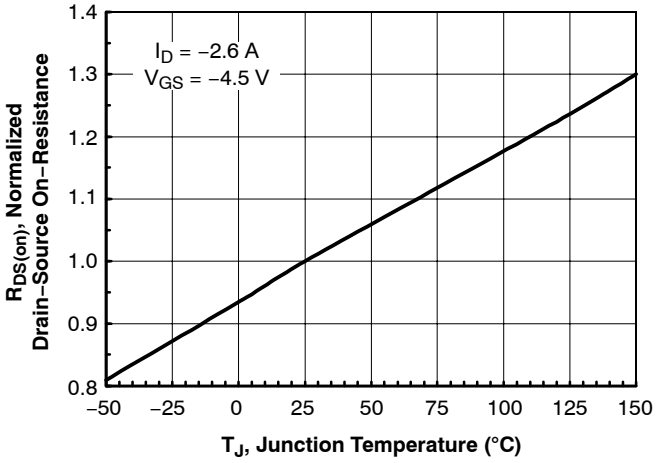

Figure 3. On−Resistance Variation with Temperature

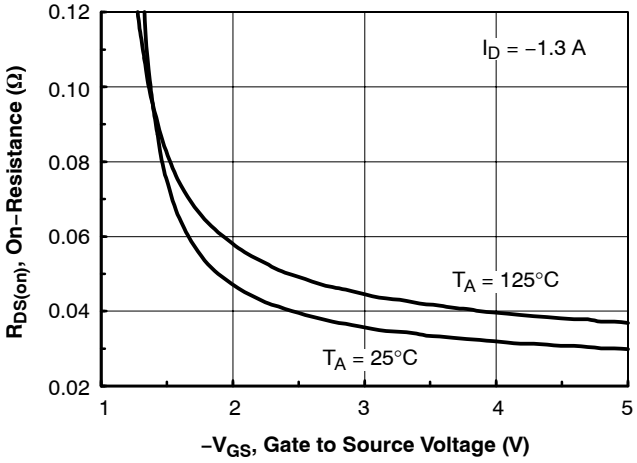

Figure 4. On−Resistance Variation with Gate−to−Source Voltage

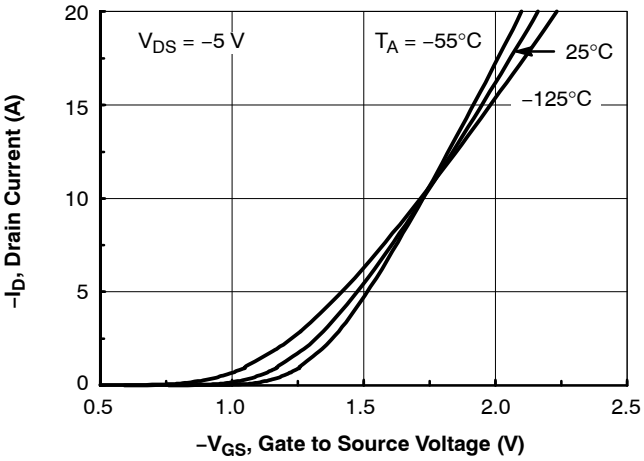

Figure 5. Transfer Characteristics

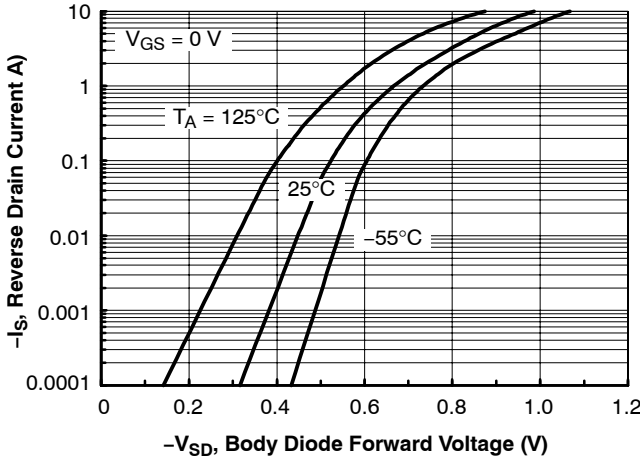

Figure 6. Body Diode Forward Voltage Variation with Source Current and Temperature

TYPICAL CHARACTERISTICS (Continued)

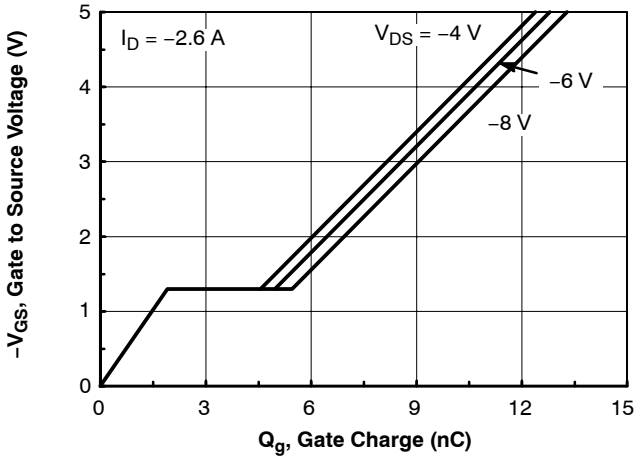

Figure 7. Gate Charge Characteristics

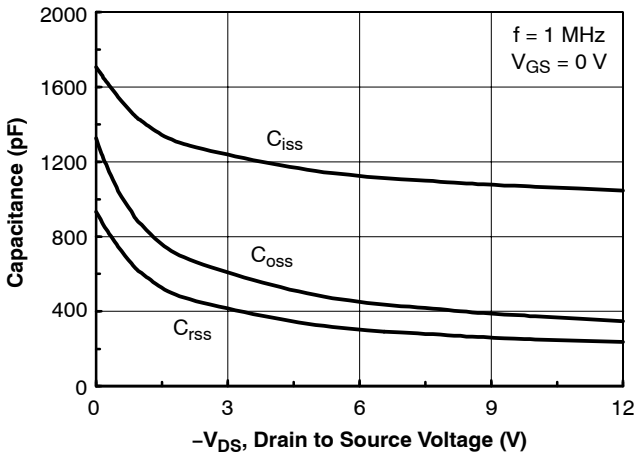

Figure 8. Capacitance Characteristics

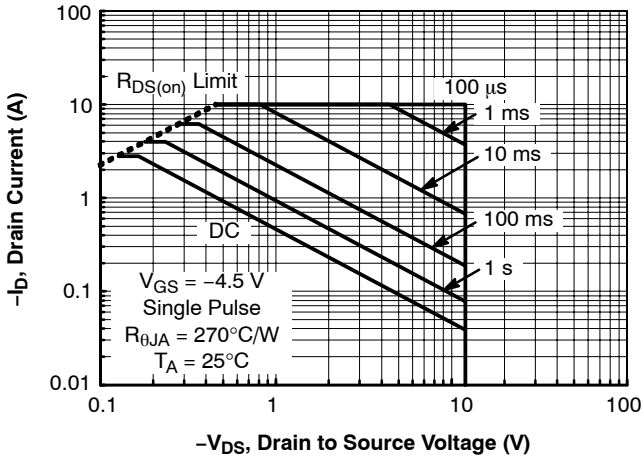

Figure 9. Maximum Safe Operating Area

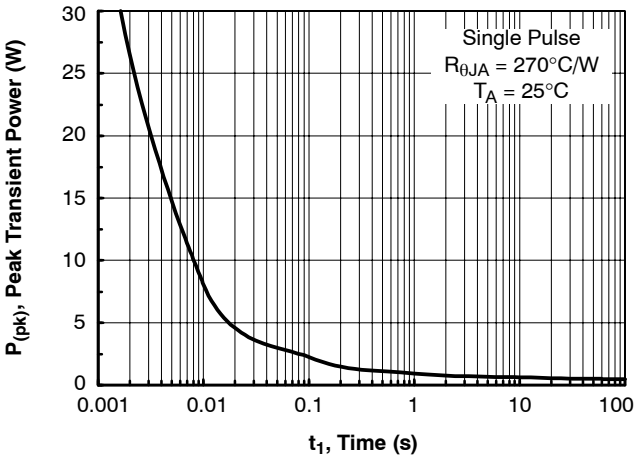

Figure 10. Single Pulse Maximum Power Dissipation

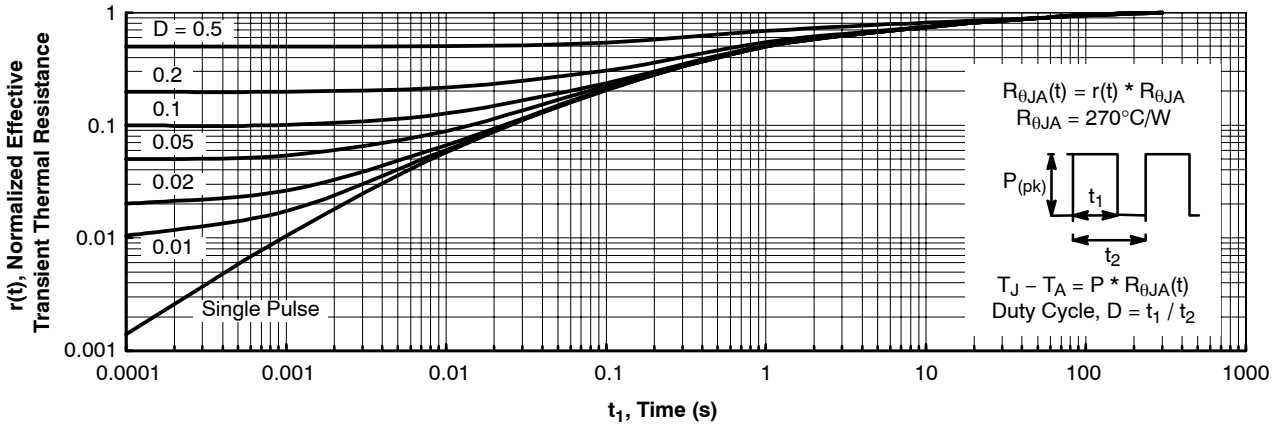

Figure 11. Transient Thermal Response Curve

Thermal characterization performed using the conditions described in Note 1b.
Transient thermal response will change depending on the circuit board design.

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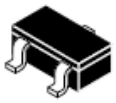
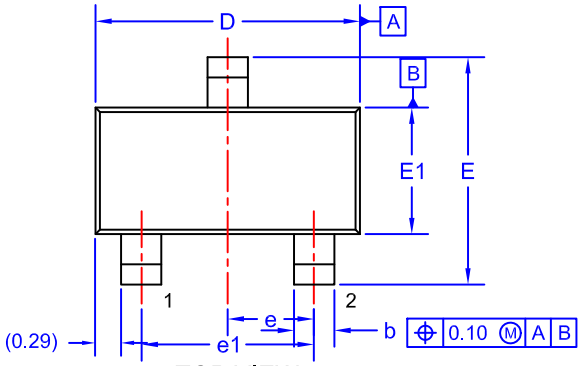


SOT−23/SUPERSOT™ −23, 3 LEAD, 1.4x2.9
CASE 527AG
ISSUE A

DATE 09 DEC 2019

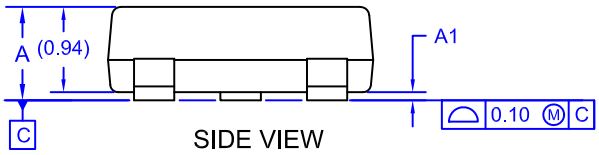


TOP VIEW

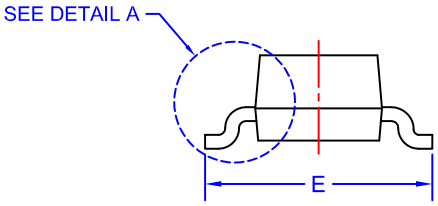
NOTES: UNLESS OTHERWISE SPECIFIED

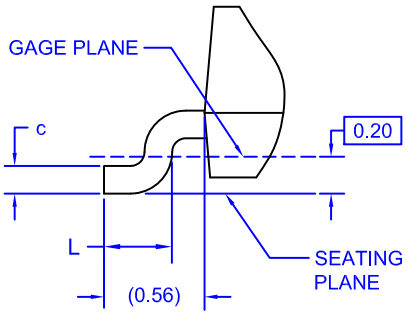
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2009.
2. ALL DIMENSIONS ARE IN MILLIMETERS.
3. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH AND TIE BAR EXTRUSIONS.

DIM	MIN.	NOM.	MAX.
A	0.85	0.95	1.12
A1	0.00	0.05	0.10
b	0.370	0.435	0.508
c	0.085	0.150	0.180
D	2.80	2.92	3.04
E	2.31	2.51	2.71
E1	1.20	1.40	1.52
e	0.95 BSC		
e1	1.90 BSC		
L	0.33	0.38	0.43

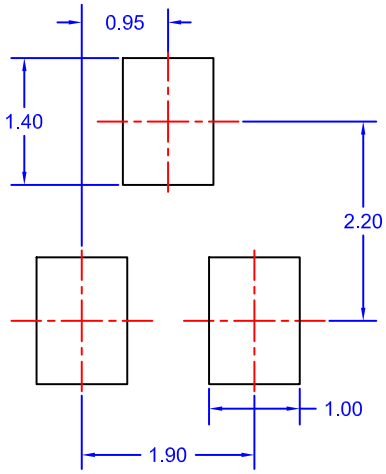


SIDE VIEW





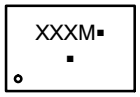
DETAIL A



LAND PATTERN RECOMMENDATION*

*FOR ADDITIONAL INFORMATION ON OUR Pb-FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE ON SEMICONDUCTOR SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



XXX = Specific Device Code
M = Month Code
▪ = Pb−Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.

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